ENGINEERING SCIENCE

Higher

Revised Sept 2019

Mechanisms and Structures

Unit 3.1

1. Friction
2. Couplings and clutches
3. Torque and mechanical power calculations
4. Drive systems





Name: _____

Class: _____

Teacher: _____

JAMES GILLESPIES'S HIGH SCHOOL

Contents

Outcomes and Assessment Standards 4
Mechanical symbols 5
What is a mechanism? 6
Why use a mechanism? 6
Motion 7
Rotary 7
Linear 7
Reciprocating 7
Oscillating 7
Forces 7
Static forces 8
Dynamic forces 8
Bending forces 8
Shear forces 8
Torsion forces 8
Compression forces 8
Tension force 8
Converting motion (transfer in motion) 9
CAMS 9
Cam motion 9
Crank & slider 10
Reciprocating motion to rotary motion 10
Rotary motion to reciprocating motion 10
Rack & pinion 10
Worm and Wheel: 10
Bevel gears: 11
Ratchet and Pawl: 11
Friction within a system 12
1. The use of lubricants 12
2. The use of bearings 12
Bearings 13
1. Split bearings 14
2. Ball-and-roller bearings 14
Friction in brakes and clutches 15

Purpose of couplings 17
Couplings: Aligned and Non-aligned 17
Aligned shafts 17
Non-aligned shafts 17
Non-aligned shafts continued 18
Clutches 18
Clutch mechanism 18
Positive clutch (sometimes called a dog clutch) 18
Friction clutch 19
Drive systems: Smooth belt 20
Different types of smooth belt 20
Drive systems: Jockey pulley 21
Stepped cone pulley system 21
Toothed belts drive systems 21
Chain drives systems 21
1. 22
2. 22
What is torque? 22
Calculation of torque 23
Basic torque equations 24
Mechanical power 26
Power transmitted by a belt drive 26
3. 27
Gearing. Exam past paper question. Q12 2019 28

Outcomes and Assessment Standards

This unit will build on skills learned in National 5 Engineering Science in order to further deepen your understanding of Mechanisms and Structures to enable you to design your own working systems.

Use traffic lighting to record your progress. This will also help with revision as you will be able to see what areas you need to work on. Your teacher will sign off this page after speaking to you about your self-evaluation:

Red: I need to work on this topic further to understand it.

Amber: I need a bit of help to fully understand this topic.

Green: I am confident in my understanding of this topic.

I can identify basic energy transformations.

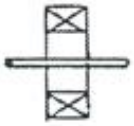
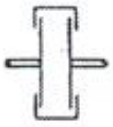
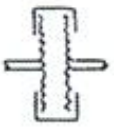
I can research and investigate a range of complex mechanisms and structures by:

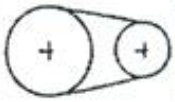
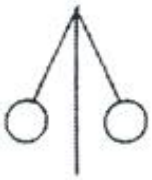
- Using the systems approach to analyse mechanisms and structures.
- Describing or producing diagrams of a range of complex structures.
- Describing or producing diagrams of a range of complex mechanisms.
- Investigating the properties of a range of materials used in mechanisms.
- I can identify basic energy transformations.
- Understand how Engineers work with friction.
- Be able to explain the need for chain and belt drive systems in particular applications.
- Be able to explain the need for bearings systems in particular applications.
- Be able to explain the need for clutch systems in particular applications.

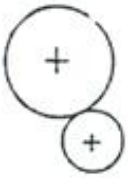
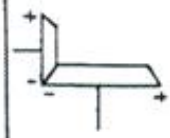
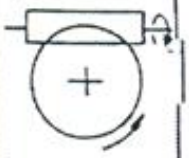
I can develop mechanical or structural solutions to solve complex problems by:

- ☐ Identifying key aspects of the problem.
- ☐ Applying knowledge and understanding of structures, materials and/ or mechanisms.
- ☐ Carrying out calculations to assist the selection of materials or component sizes.
- ☐ Designing structures and/ or mechanisms.
- ☐ Simulating or building mechanisms and/ or structures.
- ☐ Testing and evaluating solutions.
- ☐ Testing and evaluating programs against a specification.

Mechanical symbols

DESCRIPTION	SYMBOL
MOVEMENT	
Linear	
Rotational	
Linear (reciprocating)	
Rotational (reciprocating)	
Fulcrum	
Lever and Fulcrum	
Shaft	
Ball or Roller Bearing	
Coupling	
Flexible Coupling	
Cam	

DESCRIPTION	SYMBOL
Belt Drive	
Chain Drive	
Flywheel	
Governor	
Ratchet	
Pump	
Compression Spring	
Tension Spring	

DESCRIPTION	SYMBOL
Torsion Spring	
Torsion Spring	
Spur Gears	
Spur Gears (+ Movement towards observer)	
Bevel Gears	
Worm and Wheel	
Rack and Pinion	

What is a mechanism?

Mechanisms are widely used in industry and are included in a large number of devices and equipment commonly used in society. It is important, therefore, that students have a good understanding of how mechanical systems assist and aid everyday tasks. Throughout the coursework, students should gain knowledge of how mechanical systems operate and how they fit into common everyday devices. It is not only necessary to understand how the mechanical system carries out its task, but also important to know hot the systems can be altered to improve performance and efficiency. It is therefore recommended that this unit of work be completed towards the end of the course to allow students to grasp the requirements and rigour of the mechanisms calculations.

Why use a mechanism?

Mechanisms are essential in manipulating energy within moving systems. Mechanisms allow us to drive cars, use machines and even tasks as simple as open doors. The World we live in is surrounded by mechanisms so it is important that as Engineers we fully understand the place that mechanisms have in a modern society.

Mechanisms are devices that will convert one form or size of motion into another, or will convert one kind or size of force into another. A mechanism is a system of moving parts that changes an input motion and force into a desired output motion and force. Mechanisms are a large part of modern society. Most of the mechanisms are that we use everyday are so familiar that we never think twice about them, for example door handles, light switches, scissors.

Mechanisms play a vital role in industry. While many industrial processes now have electronic control systems, it is still mechanisms that provide the muscle to do the work. They provided the forces to press steel sheets into the shape of car body panels, to lift large components from place to place and to force power hacksaws to cut through thick metal bars – the list of jobs is endless. It is only by using mechanisms that industry can make products you use every day.







Some machines are easy to understand, but many are hidden away from sight behind glossy panels and covers. In the past, machines were much easier to see, as with the old steam engine, for example, but as people became more concerned about safety, it was necessary to fit guards over moving parts. Today, guards are often replaced by styled covers that make it much harder to see what is happening, but whether you can see them or not, mechanisms are still playing a vital part in everyday life.



A Mechanism changes an input motion and force into another output motion and force. All mechanisms must:

- involve some kind of motion.
- involve some kind of force.
- make a job easier to do.
- need some kind of input to make them work.
- Produce some kind of output.

Motion

To understand how a mechanism works, it is vital you know the ways it can move. There are four basic kinds of motion:

Rotary 

Turning in a circle. This is the most common type of movement, for example wheels, clock hands, CD-ROMs, compact discs or gears.

Linear 

Movement in a straight line, for example movement of a paper trimmer cutting a straight edge on paper or a lift moving between floors.

Reciprocating 

Backwards and forwards movement in a straight line, for example the needle in a sewing machine or the piston in a car engine.

Oscillating 

Swinging backwards and forwards in an arc, for example the pendulum of a clock, a playground swing or a rocking horse.

Forces

Forces affect structures in a variety of different ways depending on how they are applied to the structure. Forces can move a structure slightly or cause damage by changing the shape of the structure.

Sometimes when forces are applied to a structure, it may be almost impossible to see changes happening. For example, a bridge will sag slightly when a vehicle drives over it, but this is not visible to the human eye. Nevertheless, the vehicle causes downwards movement of the bridge structure. Loads such as vehicles on a bridge can be deemed examples of forces acting on the bridge. Forces can stop an object from moving or they can make it change direction. When a football is kicked, the forces applied from the player cause the dimensions of the ball to change on impact. It happens so quickly that it is not visible.

Forces are measured in *Newtons* and the symbol is the letter '**N**'. There are a number of different types of forces that can be applied to and which affect bodies and structures.

Static forces

When static loads or forces are applied to structures, the structures do not normally move. Normally the total downwards force comprises the weight of the structure plus the load it is carrying.

Dynamic forces

When dynamic loads or forces are applied to a structure, the structure does move and the forces applied can be varied. Dynamic forces are visually more noticeable and are produced by a variety of means and effects: machines, wind directions, people, etc.

Bending forces

Structures that carry loads across their length are subject to bending forces.

Shear forces

Shear forces can be described as tearing or cutting forces affecting structures.

Torsion forces

Torsion or torque forces have the effect of trying to turn or twist a structure or a piece of material.

Compression forces

The picture below shows the supporting columns on the new Queensferry crossing which cross over the River Forth between Fife and Edinburgh The column experience a huge downward force from the cables but they don't not disappear into the ground because the ground exerts an upwards reaction force on the column's base. The downward pressure of the weight and the upward reaction are external forces trying to squash or shorten the column. Forces that act like this are called compressive forces and the column is said to be in compression. The Queensferry crossing is called a cable stayed bridge design and is the longest 3 tower bridge in the World.



Tension force

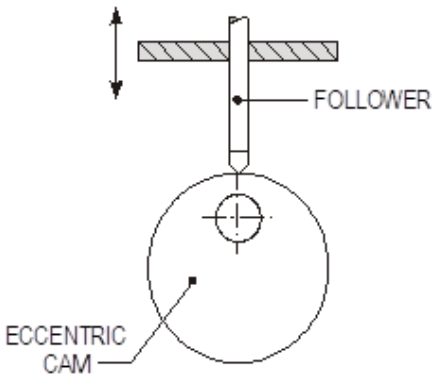
We have noted that compression occurs when things are being pushed together. The opposite of compression is 'tension' – when a structure is being pulled apart.

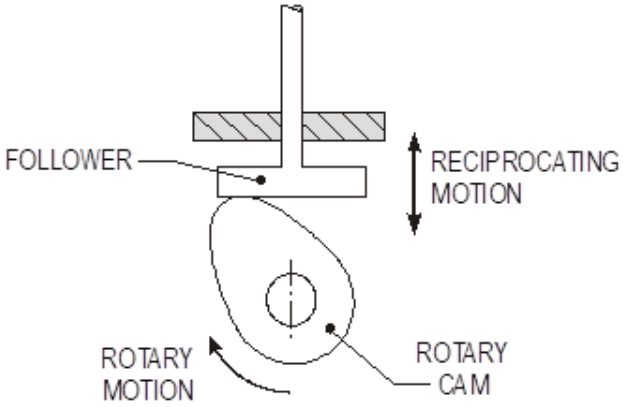
Converting motion (transfer in motion)

CAMS

A cam is a specifically shaped piece of material, which can be used to change an input rotary motion to an output motion that is oscillating or reciprocating.

The cam operates by guiding the motion of a follower held against the cam, either by its own weight or by a spring. As the cam rotates the follower moves. The way that it moves and the distance it moves depends on the cam's shape and dimensions.

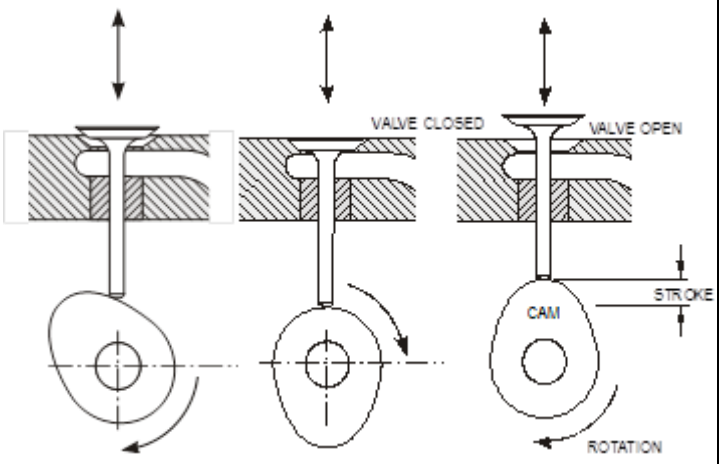


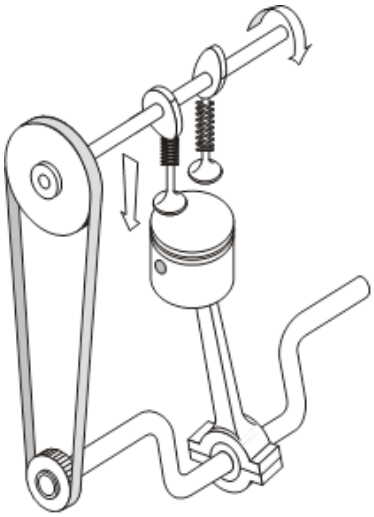


Cam motion

Pear-shaped cams are often used for controlling valves. For example, they are often used on motor-car cam shafts to operate the engine valves.

A follower controlled by a pear-shaped cam remains motionless for about half a revolution; during the other half revolution of the cam the follower rises and falls. As the pear-shaped cam is symmetrical, the rising motion is the same as the falling motion. When the follower is not moving we call this the dwell part of the cam.

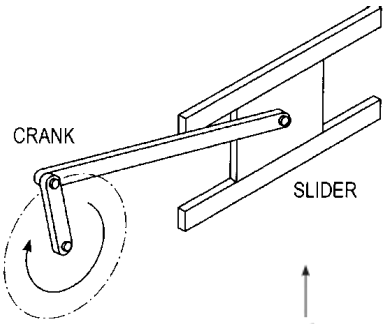




In a car engine, cams are fixed on a camshaft. As each cylinder has two valves, an inlet and an exhaust valve, there are two cams on a camshaft for each cylinder as shown.

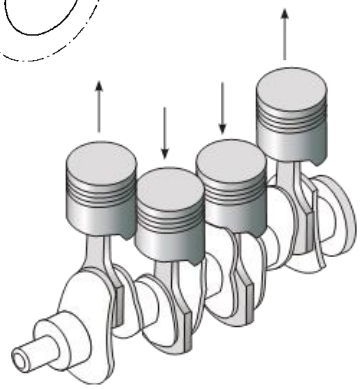
Crank & slider

Crank & slider mechanisms involve changes between rotary and reciprocating motion. The crank rotates while the slider reciprocates. The longer the crank the further the slider will move.



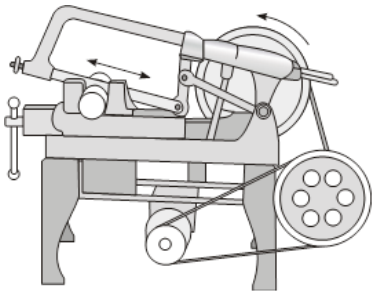
Reciprocating motion to rotary motion

Car engines use reciprocating pistons, which are connected to a crankshaft by connecting rods. As the piston moves up and down the connecting rods push the crankshaft round. Each piston moves down in turn so keeping the crankshaft turning.



Rotary motion to reciprocating motion

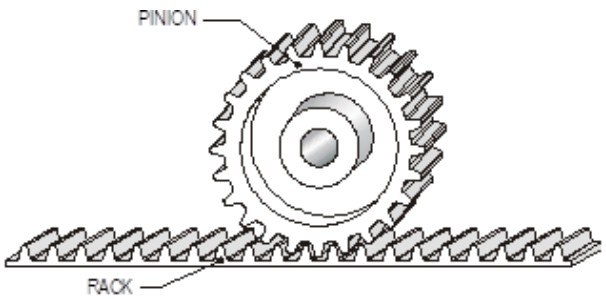
A power hacksaw uses an electric motor to power a crank, which is connected to a saw frame. The saw frame is free to slide on the arm. As the crank rotates it causes the frame to slide backwards and forwards on the arm. The longer the crank the further the saw frame will move.



Rack & pinion

A rack & pinion mechanism is used to transform rotary motion into linear motion, or linear motion into rotary motion. A round spur gear, the pinion, meshes with a rack that can be thought of as a spur gear with teeth set in a straight line.

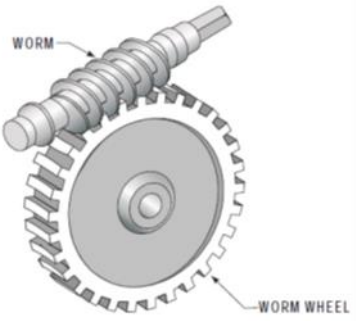
Gear wheels are normally made from metal or plastic. Plastic gears have the advantage that they are much quieter running and need less lubrication.



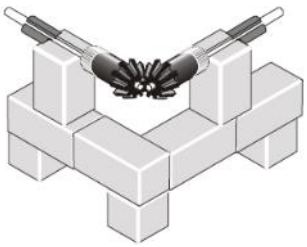
Worm and Wheel:

The worm looks like a screw thread and can be thought of as a one toothed gear is fixed to the input shaft. It meshes with the wheel which is fixed to the output shaft.

Output turns at 90 degrees to the input



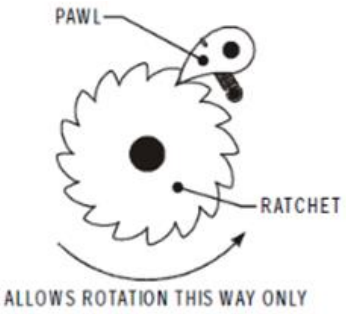
Bevel gears:

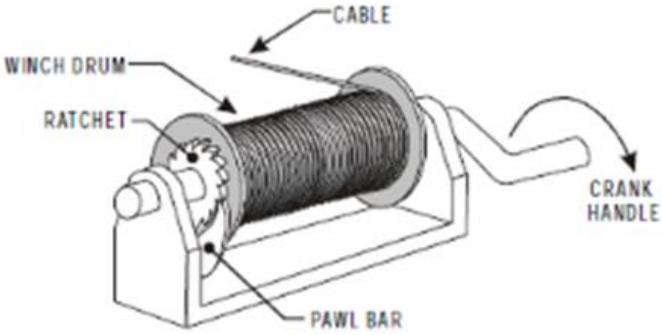


Bevel gears are used to change the direction of rotational motion through an angle. Bevel gears have their teeth machined at 45 degrees. When the meshed gears are the same size and have the same number of teeth they are called mitre gears. They will change the rotational direction through 90 degrees.

Ratchet and Pawl:

When a load is lifted by a pulley as soon as it is released you will notice how it falls straight back down due to its own weight. A mechanical system that acts like this is said to overhaul. A useful device to prevent this is a ratchet and pawl. A ratchet and pawl system allows rotation of the drum in one way only. The function of the device is a useful safety feature when used in lifting systems. Ratchet and pawl systems are also used in devices such as clockwork toys, fishing rods and socket sets.

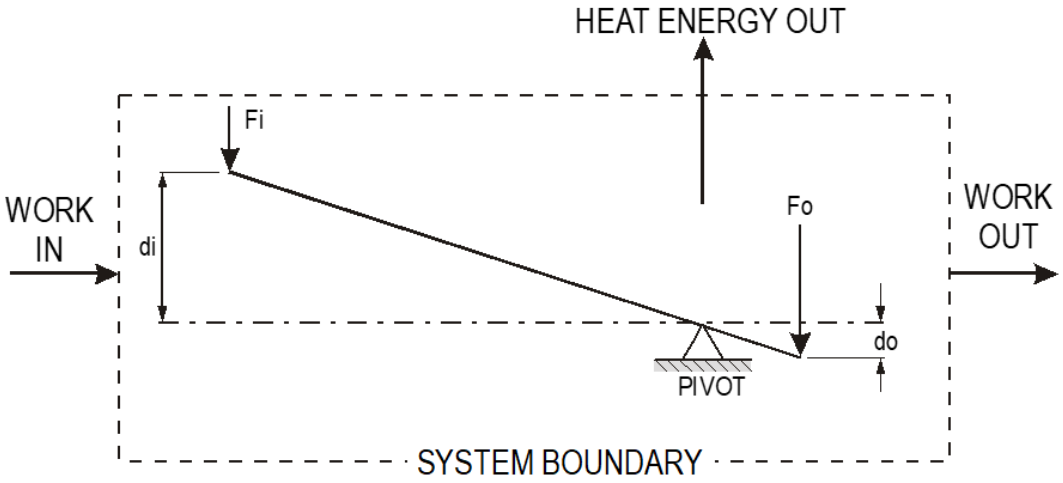




Friction

As we have seen the lever can be considered to be an ideal mechanism. It converts all the energy input to do the work at the output. In reality however this would not be the case.

Probably the lever would bend slightly as the input force was applied. Therefore the distance which this force would have to be moved would be greater thus increasing the Velocity Ratio and reducing the efficiency. Even if the lever was perfectly rigid it still could not be 100% efficient because of friction.



Friction occurs when any two surfaces come into contact. In the case of the lever, friction between the pivot and the underside of the lever will generate heat energy. This means that some of the input energy will be 'lost' at the pivot and the amount of work done at the output will be less.

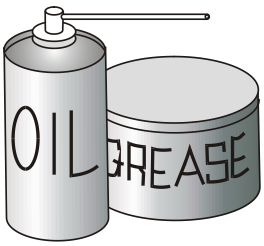
In more complex systems friction can be a major problem and engineers have developed various methods of reducing friction.

Friction is not all bad however and without friction many mechanical systems would not work.

Friction within a system

Within all drive systems, a huge amount of friction can be created, causing heat. This can cause the material of the gears/pulleys to expand, which can cause a lot of damage to the drive system. To help reduce this, two things can be done:

1. The use of lubricants



The simplest way to reduce friction is to use a lubricant such as oil or grease. By lubricating the moving parts, it allows the different parts to touch and move more easily without causing so much heat.

2. The use of bearings

Parts of mechanisms that slide over each other use flat bearings. Flat bearings tend to be made from cast iron, brass or bronze. Brass and bronze bearings, which are softer than the materials sliding through or over them, will wear. They are sometimes called wear strips. When badly worn they are replaced. Cast iron is a self-lubricating material and is very strong when compressed.

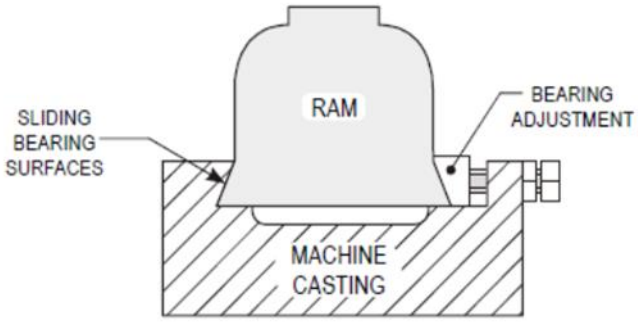
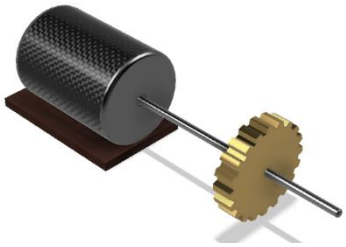
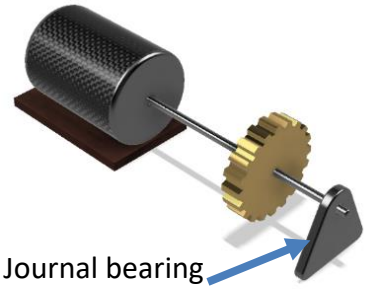


Image of a flat bearing and wear strip

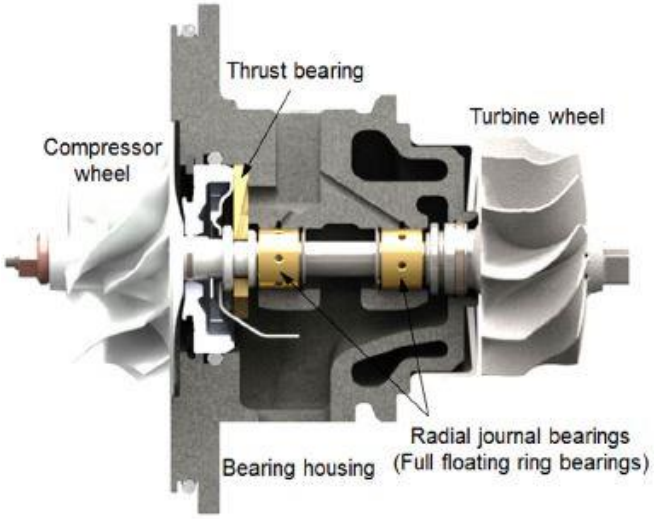
Bearings



When axles or shafts (sometimes known as journals) rotate they need to be supported. For example, if a motor with a long shaft has a gear attached then the weight of the gear is likely to make the shaft bend. This can cause significant problems in most mechanical systems as it is likely to cause damage or complete failure.



When a shaft is turned, it must be supported in some way. A bearing is the piece of kit that supports the shaft. The vertical force exerted on the bearing in this example is known as the **radial load**. The most simple example of a bearing is an axle rotating in a hole. This called a journal bearing.



If it needs to stop the axle from slipping through the bearing then it must also contain a thrust bearing.

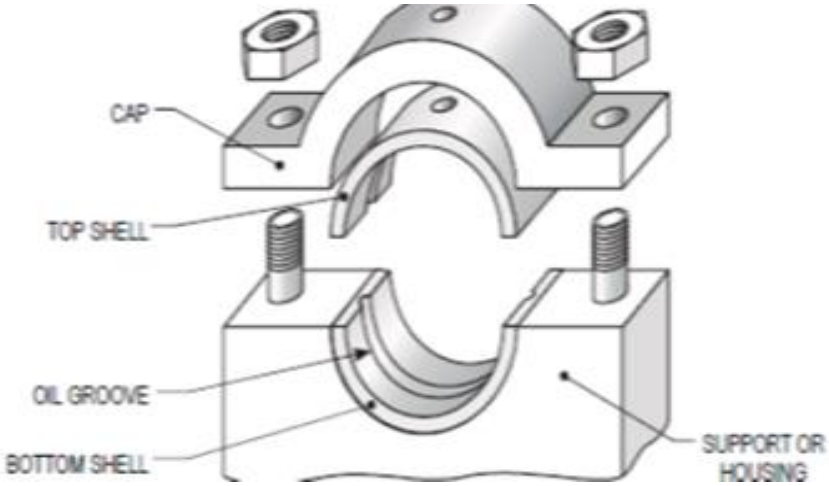
When a journal bearing has to take some large axial load for example within a turbine, it must have a shoulder to take this load but also to stop the axel from slipping through the bearing. This is called a thrust and journal bearing.

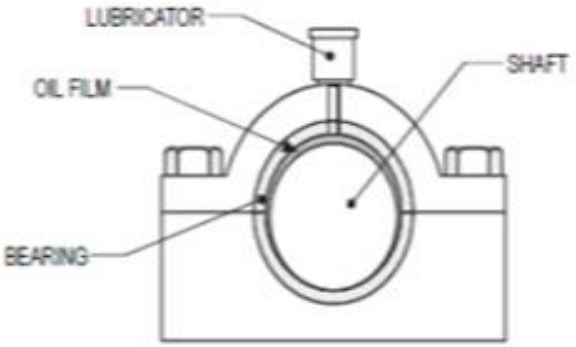
Friction opposes motion, and when a shaft is turning there is likely to be heat and wear at the supports. The amount of heat and wear due to friction will vary with the materials used, the forces involved and the speeds involved. Various types of bearing and bearing materials have been developed to reduce friction in mechanisms.

Journal bearings are made from a variety of materials: the most common are bronze and white metal. Bronze is used where slow, heavy loads are carried. White metal, an alloy of tin, copper and antimony, which is soft and melts when overheated, is used in systems with light loads. Plastic and nylon bearings are also very common.

1. Split bearings

As bearings are designed to wear, it stands to reason that they must be able to be removed and replaced. When the bearing support is at the start or end of a shaft, it is simple to remove and replace it. However, when a shaft is very long, it may be supported at several points along its length. To make it easy to remove and replace bearings, split bearings are used as shown below.





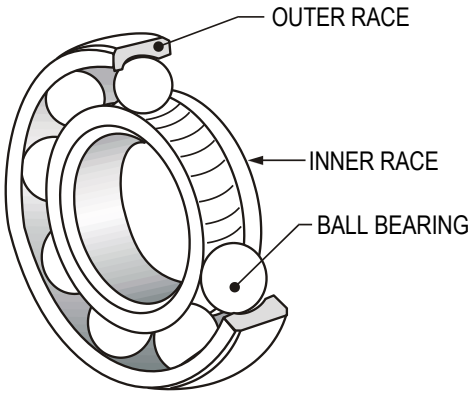
When the bearing wears, the bearing housing can be separated by removing the two nuts. The bearing shells can then be removed and replaced. Notice that the inside of the shells has a groove. This groove is normally fed by a reservoir of oil, which helps to lubricate the shaft and bearing, thus reducing friction. A car big end is a common example.

2. Ball-and-roller bearings

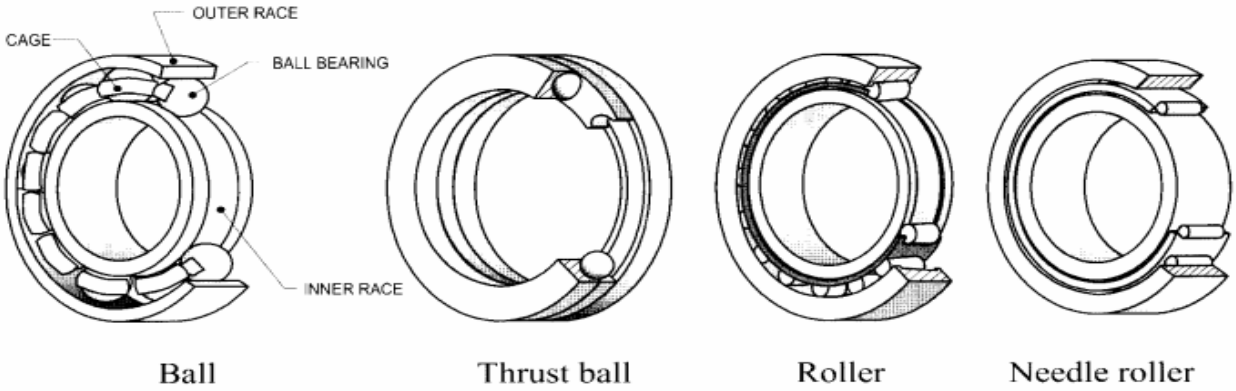
Ball and roller bearings change the action of rubbing to that of rolling. Ball and roller bearings use hardened steel balls or rollers, which rotate inside an inner and outer case. The outer case or 'race' presses into a housing; the inner race is a press fit on the shaft. These bearings are used in high-speed, high-force applications.

In moving parts such as at the hub of a wheel, ball bearings get over the problem of surfaces rubbing together.

The bearing consists of an outer and an inner 'race' which have grooves machined in them. Hardened steel spheres, or 'balls', are fitted between the outer and inner race which can then rotate freely. To work effectively the ball bearing must be well lubricated.



These are components that are design to withstand wear. Ball and roller bearings are used within high-speed and high-force systems. Bearing replace any rubbing action with a rolling action instead as rolling does not create as much friction as rubbing. Parts of mechanisms that slide over each other use flat bearings, which can be made from brass or bronze and replaced when they are worn. Split bearings are designed to wear, it stands to reason that they must be able to be removed and replaced.



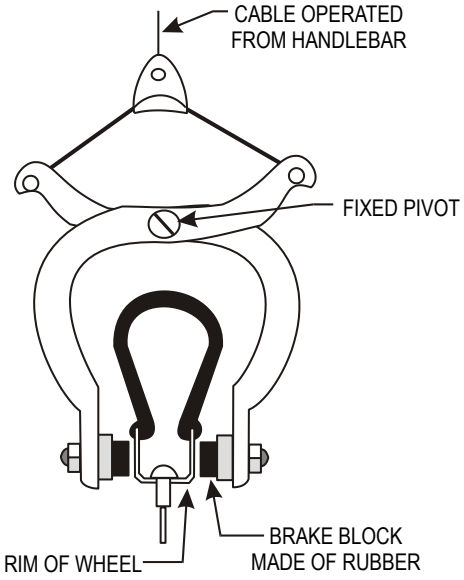
Friction in brakes and clutches

It is friction between the soles of our shoes and the pavement, which allows us to walk. If it were not for friction it would feel a bit like walking on ice and we would find it impossible to get about. Tyres on cars are designed to increase friction with the road surface to ensure that the force from the engine can be transmitted effectively.

Bell crank levers and reverse motion linkages are used to transmit the input force and motion through a right angle. By varying the lengths of the two arms of the bell crank, it is possible to use it either to magnify an input force or to magnify an input motion.

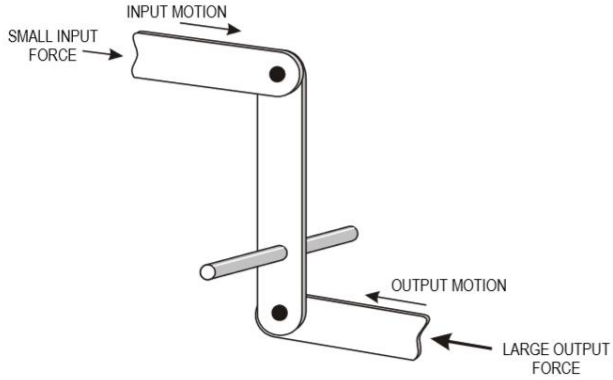
The mechanism that operates calliper brakes on bicycles uses two bell cranks to transmit braking force. These mechanisms transmit the force from the brake cable at right angles onto the wheel rim.

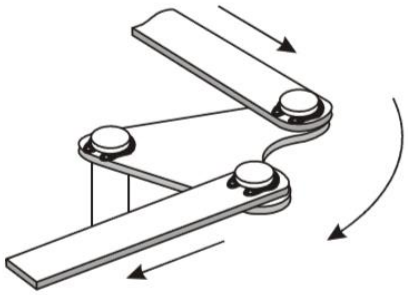
Without friction it would be impossible to stop our cars or bicycles because the brakes also rely on friction.



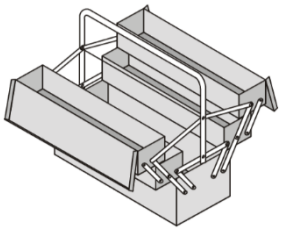
Linkages

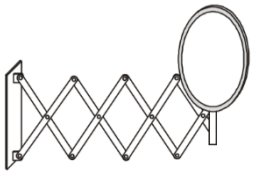
Linkages are mechanical systems which transmit force and movement. Like levers they can be used to increase the effect of an input force. Some common types of linkage are shown in the diagrams below.



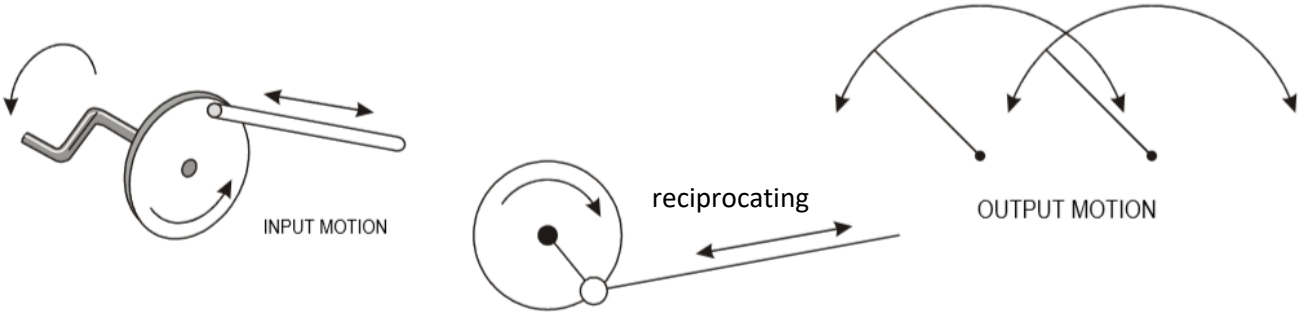


Parallel linkages are used to make two or more parts of a mechanical system move together in parallel. These systems are based on a parallelogram and are used in folding devices such as extension mirrors and tool boxes.





A common application of a parallel linkage can be seen in some car windscreen wipers. The diagram shows how this is achieved using a rotating linkage to drag a parallel linkage back and forth.



Purpose of couplings

Rotary machines employ a variety of methods of transmitting motion from one part of a machine to another. The motion of often transmitted through shafts, which are round metal rods. Often these shafts must be connected together to transmit the motion.

Shafts are joined using a device called a coupling. In small models, such as those used in schools, simple sleeves or tubes of plastic use friction to drive two shafts, which are pressed into the sleeve. Stronger couplings are required for industrial-sized machines.

Couplings: Aligned and Non-aligned

Aligned shafts

Where shafts are in line with each other they are joined either with a flanged coupling or a muff coupling. All couplings must be 'keyed' to the shafts they are joining to give a positive drive. Figure 1 shows a flange coupling then a muff coupling.

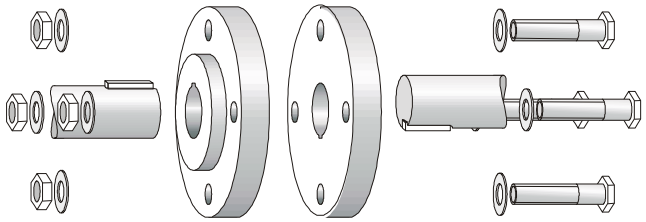


Figure 1(a): flange coupling

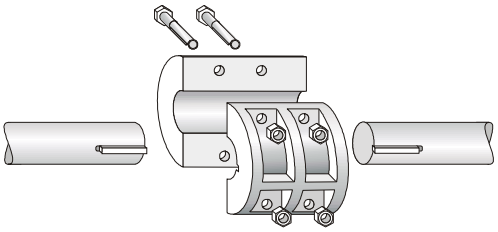


Figure 1(b): muff coupling

Non-aligned shafts

Where shafts meet at a slight angle, some method of compensating for misalignment must be used. Where the misalignment is small, a flexible coupling (flexi-coupling), using either rubber or a mixture of rubber and steel, is used. The rubber is flexible enough to compensate for small changes in angle (figure 2).

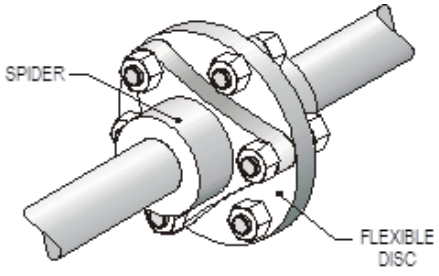


Figure 2: flexi-coupling

Non-aligned shafts continued

When the alignment is more than a few degrees out, a universal joint is used. A universal joint can transmit motion through an angle of 20 degrees. Figure 3 shows Hooke's universal joint. The two yokes are free to pivot on the central 'spider'. Modern universal joints use needle roller-bearings between the spider and the yokes.

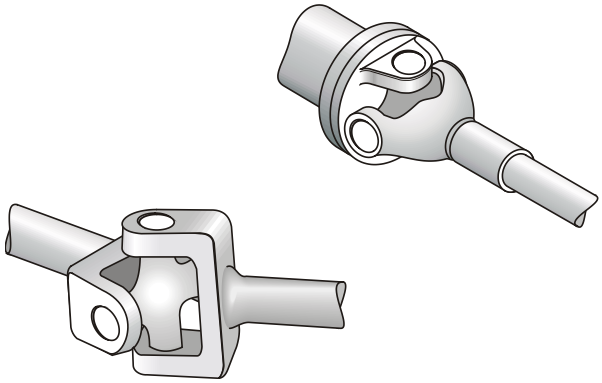


Figure 3: universal joints

Clutches

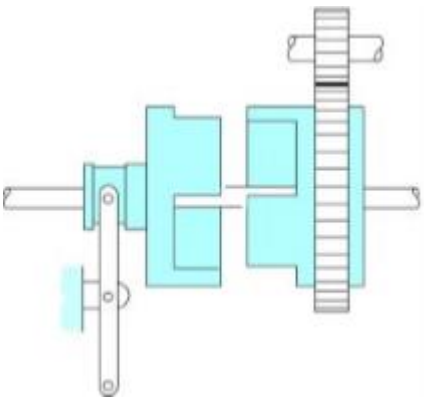
Clutch mechanism

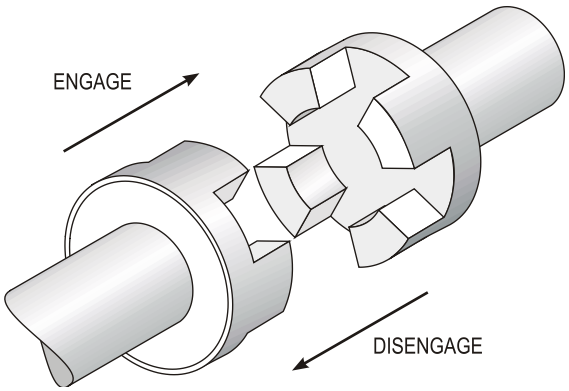
A clutch is a device that allows two rotating shafts to be connected and disconnected. It would be very inconvenient if a driver had to stop and restart the car engine every time she stopped the car at a traffic light or road junction. A driver needs a mechanism that allows her to disconnect the engine from the shaft that drives the wheels, so that she can stop the wheels without having to stop the engine. This is what the clutch does. There are two main kinds of clutch.

Positive clutch (sometimes called a dog clutch)

A dog clutch is a positive clutch. This has four interlocking blocks (dogs) on one shaft that can be interlocked with four dogs on the other shaft.

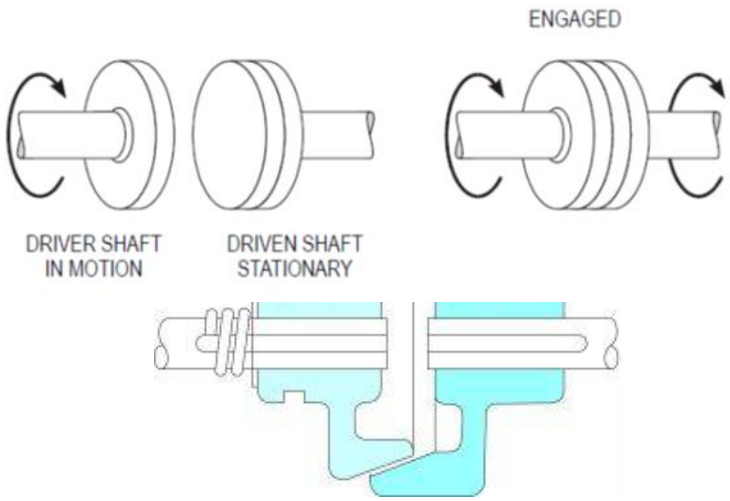
When the clutch is engaged, the two dogs are interlocked and the drive shaft rotates the driven shaft. When the clutch is disengaged, the two shafts are separated. In clutch systems, the two shafts must be carefully aligned. Positive-drive clutches require the drive shaft to be stationary when the two clutch plates are brought together. The term 'engaging the clutch' means bringing the two plates together so that the two shafts are connected.





Friction clutch

Friction clutches can be engaged and disengaged while both shafts are still turning. Friction clutches rely on the friction between the plates to transmit the power from one shaft to another as shown below.



Advantages and disadvantages of clutches

Advantages	Disadvantages
Allows user to:	May be complex with parts that can fail.
Connect/disconnect drive.	Require regular maintenance.
Make adjustments.	Expensive due to replacing parts as they wear.
Change gears.	Can slip and waste energy.
Change speed.	

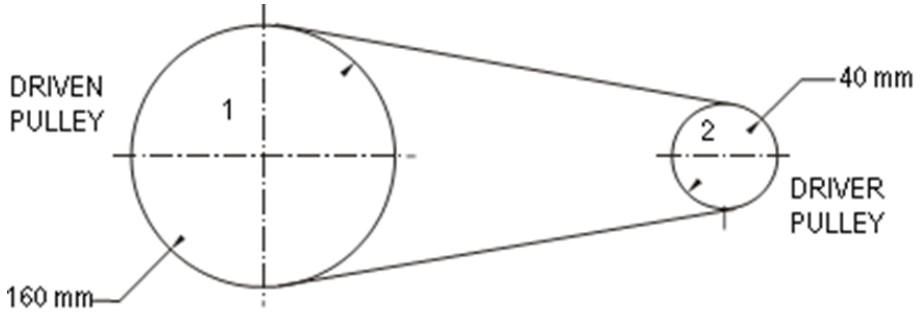
Task *Couplings*

1) Why are couplings used in mechanical systems?
2) What term is used in making sure that a coupling has a positive drive?
3) When alignment is a problem in shafts what mechanical device can be fitted?
4) Why can friction be a problem in mechanical devices?
5) What consequences may occur if friction is not overcome?
6) Thrust bearings are often used in rotating systems. What is the main advantage in using this type of bearing?
7) If a bearing is required in the middle of a long shaft, how can the problem of changing it be overcome?
8) How does lubrication work within the bearing housing?
9) Some shafts require running at high speeds with limited friction. How can this be done?
10) Explain the terms ‘dog clutch’, ‘friction clutch’ and ‘multi-plate clutch’

Drive systems: Smooth belt

Many mechanisms make use of rotary motion, often provided by someone turning a handle or by an electric motor. But to be useful, this rotary motion has to be frequently transmitted from one part of a mechanism to another, often with a change of speed by varying the diameter of the driver pulley and the driven pulley. While gears can be connected together in a simple gear train, if too many gears are connected then it can result in large efficiency losses through friction.

There are two simple means of transmitting rotary motion over relatively large distances; belt or chain. The tensioned belt transmits the rotary motion from pulley 2 to pulley 1 but also a change of speed is achieved. The belt is tightened or tensioned by pulling one of the pulleys out and locking it in place. Pulleys are thin metal discs with a groove cut into the circumference of the disc.

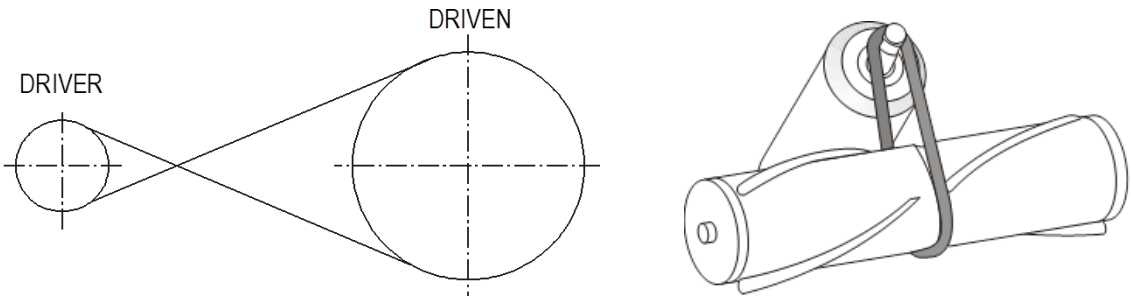


Different types of smooth belt

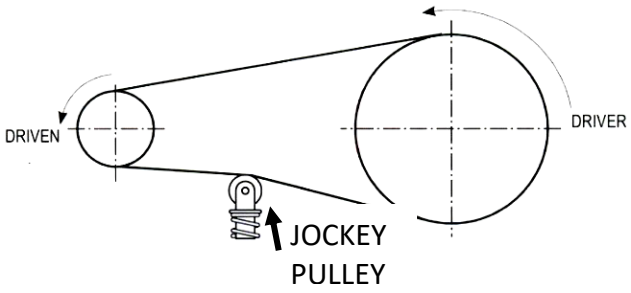
Belt drives are used in a wide variety of situations. They are made from a composite of two materials, rubber and string. The string helps to prevent the rubber from stretching too much. Drive belts are inexpensive to produce. They are easy to replace and need little maintenance, as they do not require lubrication. Belt drive systems are found in many household machines such as washing machine, tumble driers and vaccum cleaners.

A ‘V’ or ‘Vee’ belt is angled to give better grip which prevents the belt from slipping when compared against a flat belt design. A change in direction can also be achieved if the belt is crossed over as shown below. V-belts can also be used to transfer motion 90 Degree of the driven from the driver.



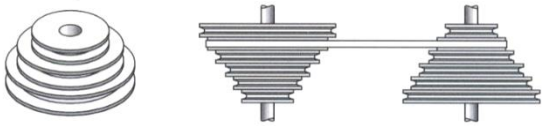


Drive systems: Jockey pulley



The advantage of a belt drive system is that it will absorb shock as it is **allowed** to slip. It is used in machines where you would not want the motor to stop or cease. For instance, if a belt drive is used to transmit the power from a motorcycle engine to the rear wheel and the biker tries to 'wheelie', the smooth belt allows slippage, preventing damage to the engine. However, excessive slipping will create inefficiency in the system. At the same time, if the belt is too tight the pulley bearings will get damaged. One way of keeping the correct tension is to use a spring loaded jockey pulley.

Stepped cone pulley system

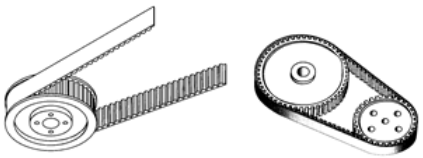


A variety of output speeds and torques can be achieved using this system. The drive motor is attached to one set of pulleys and the drive belt can be moved between various pairs of pulleys to give a selection of speeds.

Toothed belts drive systems



Toothed belt

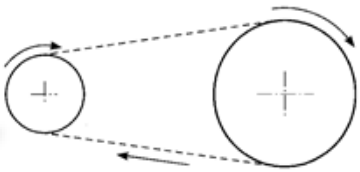


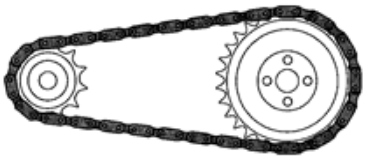
Belt drives tend to use their ability to slip to their advantage, however where slippage would damage a mechanism toothed belts have been developed that retain the advantages of normal belts but **do not slip**. Many cars have toothed belts (for example timing belts) to control the opening and closing of the inlet and outlet valves in the car engine. If the belt slipped, the pistons would collide with the valves, damaging the engine. These belt drives are quiet, require little maintenance and are easily changed if required.

Chain drives systems

Where large forces have to be transmitted and there can be **no slippage allowed**, chain drives are used. Instead of a pulley, a toothed wheel known as a sprocket is used to drive a chain. Different sprockets sizes will vary speed.

Notice: Chain system diagrams are drawn with **dashed** lines.





The best example of a chain drive system is the one found on bicycles. When cycling, if you want to go suddenly faster, you stand up and put extra weight into the pedals, which will in turn transfer greater forces to the back wheel and make you go faster or easier. Chain systems transfer greater forces as they are much stronger, unless the chain or sprockets is damaged. However, they do have to be oiled regularly, and both the chain and sprockets are prone to wear. They are also more expensive and much noisier than belt drives.

Task: Drive systems

1. Many machines and mechanisms use belts and pulleys to transmit rotary motion. Write down any machines or mechanisms that you know of which use belts and pulleys.

_____ _____ _____

_____ _____ _____

_____ _____ _____

2. ***Task 2 Drive systems:***

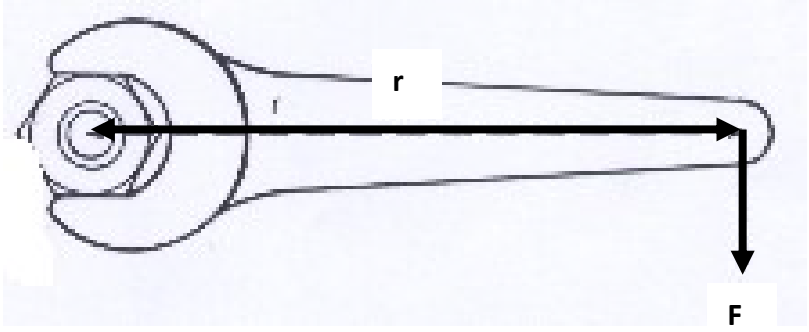
Draw a universal systems diagram for one of your above answers.

What is torque?

A force is needed to move an object, but it must be applied in a way that creates a twisting or turning action. Torque, T, is the rotational version of force, and results from the application of one or more forces. Torque is specified relative to a chosen rotation axis or pivot.

In simple terms, torque is the amount a car's engine can apply a 'rotational force'. It differs from horsepower as it refers to the amount of work an engine can exert, while horsepower defines how quickly that work can be delivered.

Try loosening a nut by gripping the end and turning it. Then, use a spanner. You should be able to feel a difference. Adding the spanner makes it easier to loosen the nut. This is because the distance (r) between the centre of the axle and the end of the spanner is acting like a lever. It increases the turning force (F) that you are applying. So, the farther away from the centre of the circle you apply the force, the greater the torque will be.



Calculation of torque

Anyone who has ever opened a door has an intuitive understanding of torque. When a person opens a door, they push on the side of the door **farthest** from the hinges. Pushing on the side **closest** to the hinges requires considerably more force. Although the work done is the same in both cases (the larger force would be applied over a smaller distance) people generally prefer to apply less force, hence the usual location of the door handle.

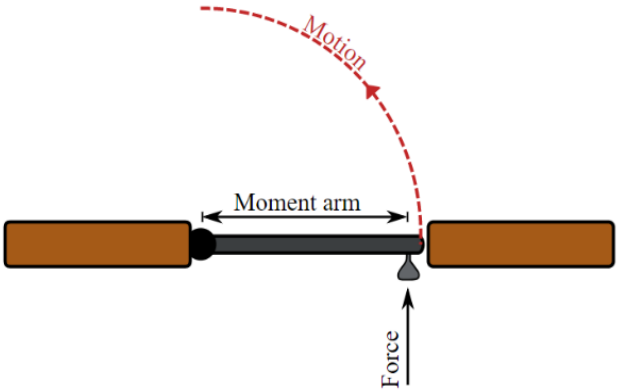


Figure 1: Opening door with maximum torque

There are two things which we use to measure 'Work' done.

Work = Force x distance

Torque or 'work' done is dependent upon:

- **F is the applied Force** and measured in Newtons (N).
- The distance from the rotation axis to the force application point and is measured in metres (m)

Therefore the unit of 'work' done or Torque will be Newton-metres (Nm). This unit of work is also given the special name, Joule (J).

For example: 1 Nm or 1 Joule of '*Work*' is generated when a Force of 1 Newton moves the body on which it is acting 1 metre in the direction of the force.

$$T = F \cdot r \sin(\vartheta)$$

where 'r' is the length of the moment arm and ϑ is the angle between the force vector and the moment arm. In the case of the door shown in Figure 1, the force is at right angles (90°) to the moment arm, so the Sin ϑ term becomes 1 and simplifies the equation to:

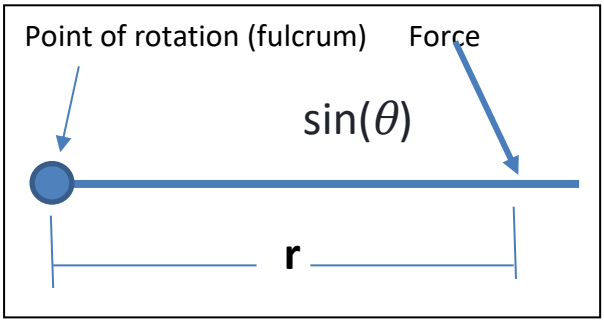


Figure 2: The orientation of the force.

T = Force x radius

T = F.r

Basic torque equations



Let's say we're using a 0.5m long wrench to tighten a wheel nut, and we need to lean on the far end of the wrench with a force of 50 Newtons to do it up tightly. Simply multiplying the two numbers gives us the required torque figure in Newton metres:

Torque = Force x distance (r)

Torque = 50 (N) x 0.5 (m)

Torque = 25Nm or 25J

Example 1:

How much torque is required to tighten the nut if the force required is 45 N and the radius of the tool is 200 mm

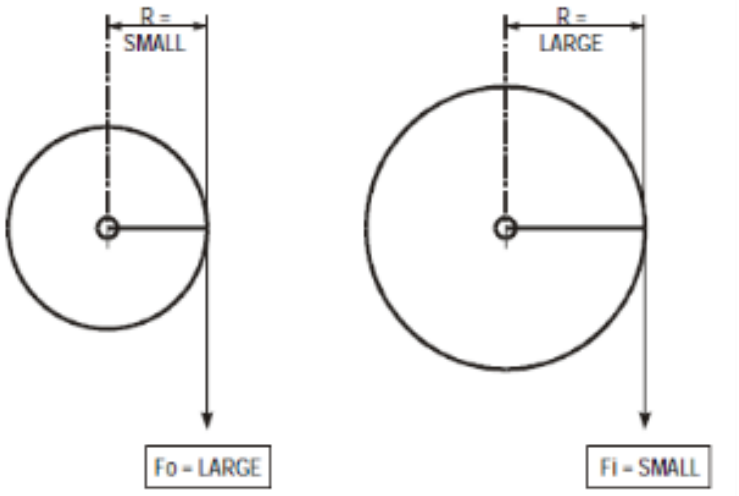
Torque = force x radius

= 45 N x 0.2m

= 9 Nm

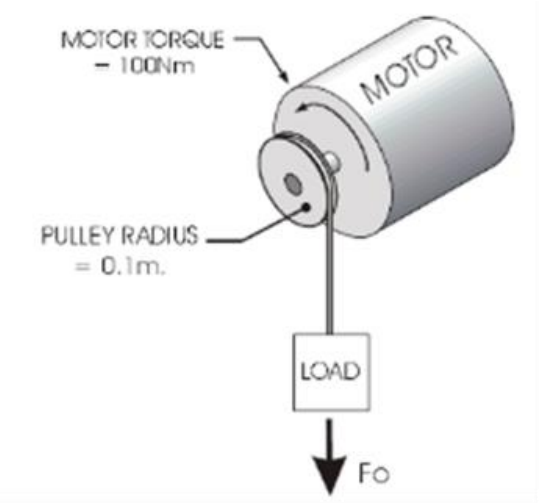
To understand Torque further, we can consider a motor such as shown in the next example. The motors ability to transmit a force through a pulley to lift a load will depend on the torque which the motor can produce and the radius of the pulley.

As the motor torque remains constant this means that the motor can be used to lift a small load quickly using a large diameter pulley or a large load slowly using a small diameter pulley.



Example 2:

The motor shown can produce a torque of 100Nm and the radius of the pulley is 0.1m. Calculate the load which can be raised by the motor.



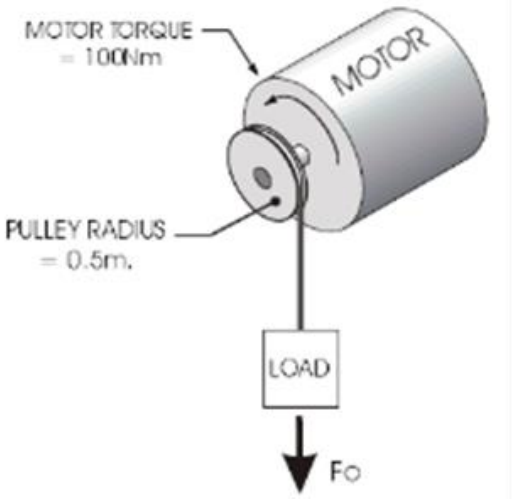
T = F x r

F = T/r

F = 100/0.1

F = 1000N

If we increase the size of the pulley to 0.5m, then a smaller load will be lifted because the motor produces the same torque.



T = F x r

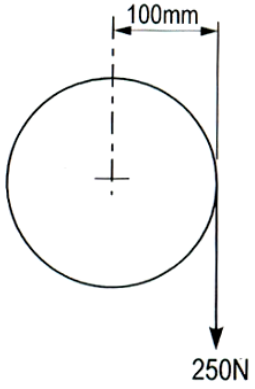
F = T/r

F = 100/0.5

F = 200N

Example 3:

A belt drives a pulley, shown opposite, with a shaft of diameter 'd' of 200mm. If the effective belt tension tending to turn the pulley is 250N, find the **Work done per revolution**. Let the applied force turn one revolution. Therefore 'F' will have travelled once round the circumference of the shaft.



Distance moved by:	F	= circumference of shaft = 2πr
Therefore Work done:	W_d	= Force x distance
		= F x 2πr
		=2πFr
But torque (T)	T	= Fr

Therefore the Work done per (one) revolution is: W_d = 2πT

For any number of revolutions ('n'), the total Work done would be calculated by:

Work done = 2πnT

Mechanical power

The mechanical power is the power created as the system turns. This differs from torque as, although both forces are calculated from rotation, for power to be produced, movement over a distance must happen. A system with no rotation can deliver torque – like an electric motor – but since no distance is moved by force, no power will be produced. Mechanical power is calculated using one of the following equations.

$$\text{Power} = \text{Force} \times \text{velocity}$$

$$P = Fv$$

- Force is measured in Newtons (N)
- Velocity (or speed) is measured in metres per second (ms^{-1})
- The unit for measuring power is Watts (W)

Or

$$P = 2\pi nT$$

Power = 2 x 3.14 x rotational speed per second x Torque

- n = is the rotational speed or revolutions per second (rev sec^{-1})
- π is 3.14
- Torque is measured in Newton metres (Nm)
- The unit for measuring power is Watts (W)

Power transmitted by a belt drive

Teacher worked example: The effective pull on a belt drive is 420 N when driving a 500 mm diameter pulley. The speed of the rotation is 220 revolutions **per minute**. Calculate the Power (Watts W).

Note the units of measurements:

When a force, 'P' newtons, acts at the rim of a pulley, of 'r' metres radius, revolving at 'n' revolutions per second, the power or work done per second is given by P = 2prn.
Power = force (P newtons) x circumference (2pr) x revolutions/s (n)
Thus power, or Work done = torque (Pr) x angle rotated through/s (2pn) = 2pnT

The effective driving torque = force x radius = (T1 –T2) diameter (d) 2

T1 is the tension on the tight side and T2 is the tension on the slack side.

Therefore power transmitted = pdn (T1 –T2)
n = 220 rpm = 220/60 = 3.7 revs/s
force = 420N
radius = 250mm = 0.25m

Power = $2\pi nT$
= 2 x 3.14 x 3.7 x (420 x 0.25)
= 2417W

3. Torque and power questions

1. Calculate the power transferred if a 230mm diameter pulley wheel revolves at 25 revolutions per second. The pulley tension force is 200N.
2. A shaft transmits 18 kW when rotating at 200 rpm. What is the torque in the shaft?
3. A railway traction motor develops 150 kW when the train moves along the track. The rail wheel rotates at 1500 rpm. Calculate the torque in the driving axle.
4. An electric motor exerts a torque of 23 Nm and rotates at 2800 rpm. Find the power of the motor.
5. The effective pull on a belt is 360 N when driving a 400mm diameter pulley. The speed of rotation is 250 rpm. Calculate the power.
6. During a machining test on a lathe, the tangential force on the cutting tool was found to be 220 N. If the work-piece diameter was 120 mm, what was the torque on the lathe spindle?
7. Calculate the power transferred if a 230 mm diameter pulley wheel revolves at 25 revolutions per second. The pulley has one belt and the tension in the tight side of the belt is 436 N, while in the slack side it is 186 N.
8. A shaft transmits 18 kW when rotating at 200 rpm. What is the torque in the shaft?
9. A railway traction motor develops 150 kW when the train moves along the track. The rail wheel rotates at 1500 rpm. Find the torque in the driving axle.
10. An electric motor exerts a torque of 23 Nm and rotates at 2800 rpm. Find the power of the motor.
11. The effective pull on a belt is 360 N when driving a 400 mm diameter pulley. The speed of rotation is 250 rpm. Calculate:

    a) the power without slip.

    b) the power with three per cent slip.
12. During a machining test on a lathe, the tangential force on the cutting tool was found to be 220 N. If the work-piece diameter was 120 mm, what was the torque on the lathe spindle?
13. Determine the torque when a pulley wheel of diameter 300 mm has a force of 80 N applied at the rim.
14. Determine the force applied tangentially to a bar of a screw jack at a radius of 800 mm, if the torque required is 600 N/m.
15. A constant force of 150 N is applied at a tangent to a wheel of diameter 140 mm. Determine the work done in 12 revolutions of the wheel. Your answer should be in joules and watt-hours.

16. Calculate the torque developed by a motor whose spindle is rotating at 1000 rpm and developing a power of 2.5 kW.

17. An electric motor develops a power of 3.75 kW and a torque of 12.5 N/m. Determine the speed of rotation of the motor in rpm.

18. A pulley is 600 mm in diameter and the difference in tension on the two sides of the driving belt is 1.5 kN. If the speed of the pulley is 500 rpm, determine:

    a) the torque developed.

    b) the work done in three minutes.

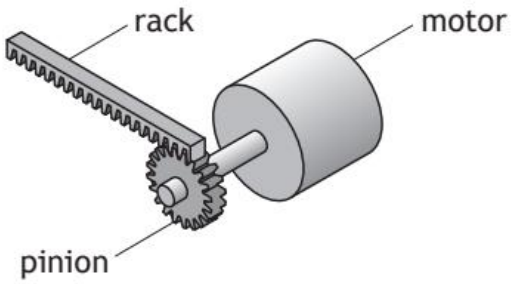
19. A 15 kW motor is driving a shaft at 1150 rpm by means of pulley wheels and a belt. The tensions in the belt on each side of the driver pulley are 400 N and 50 N and the diameters of the driver and driven pulley wheels are 500 mm and 750 mm respectively. Calculate:

    a) the power output.

    b) the efficiency of the motor.

Gearing. Exam past paper question. Q12 2019

Part of the design involves rotary motion, from a motor, transforming into linear motion.



The pinion gear has 24 teeth and the pitch of the teeth on the rack is 3·0 mm.

The rack is required to move 2·75 m in three seconds.

(d) Calculate the required speed of the motor. **3**